Ex No.	1	**Install Hadoop and configure Hadoop Distributed File System (HDFS) in standalone or pseudo-distributed mode.**
Date		

AIM:

To perform setting up and installing Hadoop in its three operating modes:

- Standalone
- Pseudo Distributed
- Fully Distributed

DESCRIPTION:

Hadoop is written in Java, so you will need to have Java installed on your machine, version 6 or later. Sun's JDK is the one most widely used with Hadoop, although others have been reported to work.

Hadoop runs on Unix and on Windows. Linux is the only supported production platform, but other flavors of Unix (including Mac OS X) can be used to run Hadoop for development. Windows is only supported as a development platform, and additionally requires Cygwin to run. During the Cygwin installation process, you should include the openssh package if you plan to run Hadoop in pseudo- distributed mode.

ALGORITHM:

STEPS INVOLVED IN INSTALLING HADOOP IN STANDALONE MODE:

1. Command for installing ssh is **“sudo apt-get install ssh”**.
2. Command for key generation is **ssh-keygen –t rsa –P “ ”.**
3. Store the key into rsa.pub by using the command **cat $HOME/.ssh/id_rsa.pub** >> **$HOME/.ssh/authorized_keys**
4. Extract the java by using the command **tar xvfz jdk-8u60-linux-i586.tar.gz.**
5. Extract the eclipse by using the command **tar xvfz eclipse-jee-mars-R-linux-gtk.tar.gz**
6. Extract the hadoop by using the command **tar xvfz hadoop-2.7.1.tar.gz**
7. Extract the hadoop by using the command **tar xvfz hadoop-2.7.1.tar.gz**
8. Move the java to **/usr/lib/jvm/** and eclipse to **/opt/** paths. Configure the java path

in the eclipse.ini file

9. Export java path and hadoop path in ./bashrc
10. Check the installation successful or not by checking the java version and hadoop version
11. Check the hadoop instance in standalone mode working correctly or not by using an implicit hadoop jar file named as word count.
12. If the word count is displayed correctly in part-r-00000 file it means that standalone mode is installed successfully.

ALGORITHM:

STEPS INVOLVED IN INSTALLING HADOOP IN PSEUDO DISTRIBUTED MODE:

1. In order install pseudo distributed mode we need to configure the hadoop configuration files resides in the directory/home/lendi/hadoop-2.7.1/etc/hadoop.
2. First configure the hadoop-env.sh file by changing the java path.
3. Configure the core-site.xml which contains a property tag, it contains name and value. Name as fs.defaultFS and value as hdfs://localhost:9000
4. Configure hdfs-site.xml.
5. Configure yarn-site.xml.
6. Configure mapred-site.xml before configure the copy mapred-site.xml.template tomapred- site.xml.
7. Now format the name node by using command hdfs namenode –format.
8. Type the command start-dfs.sh,start-yarn.sh means that starts the daemons like Name Node, Data Node,SecondaryNameNode ,Resource Manager,Node Manager.
9. Run JPS which views all daemons. Create a directory in the hadoop by using command hdfs dfs –mkdr /csedir and enter some data into lendi.txt using command nano lendi.txt and copy from local directory to hadoop using command hdfs dfs –copy FromLocal lendi.txt /csedir/and run sample jar file word count to check whether pseudo distributed mode is working or not.
10. Display the contents of file by using command hdfs dfs –cat /newdir/part-r-00000.

FULLY DISTRIBUTED MODE:

INSTALLATION: ALGORITHM:

1. Stop all single node clusters

   $stop-all.sh

2. Decide one as Name Node (Master) and remaining as Data Nodes (Slaves).

3. Copy public key to all three hosts to get a password less SSH access

   $ssh-copy-id –I $HOME/.ssh/id_rsa.pub lendi@l5sys24

4. Configure all Configuration files, to name Master and Slave Nodes.

   $cd $HADOOP_HOME/etc/hadoop

   $nano core-site.xml

   $ nano hdfs-site.xml

5. Add hostnames to file slaves and save it.

   $ nano slaves

6. Configure $ nano yarn-site.xml

7. Do in Master Node

   $ hdfs namenode –format

   $ start-dfs.sh

   $start-yarn.sh

8. Format Name Node
9. Daemons Starting in Master and Slave Nodes
10. End

INPUT:

ubuntu @localhost> jps

OUTPUT:

Data node, name nodem Secondary name node,

NodeManager, Resource Manager

RESULT:

Hence Installation of Hadoop in different modes was studied.

Ex No.	2	Demonstrate basic HDFS commands for file creation, upload, download, replication, and deletion operations.
Date		

AIM:

To implement the following file management tasks in Hadoop:

- Adding files and directories
- Retrieving files
- Deleting Files

DESCRIPTION:

HDFS is a scalable distributed file system designed to scale to peta bytes of data while running on top of the underlying file system of the operating system. HDFS keeps track of where the data resides in a network by associating the name of its rack (or network switch) with the dataset. This allows Hadoop to efficiently schedule tasks to those nodes that contain data, or which are nearest to it, optimizing bandwidth utilization. Hadoop provides a set of command line utilities that work similarly to the Linux file commands, and serve as your primary interface with HDFS. We are going to have a look into HDFS by interacting with it from the command line. We will take a look at the most common file management tasks in Hadoop, which include:

- Adding files and directories to HDFS
- Retrieving files from HDFS to local file
- system Deleting files from HDFS

ALGORITHM:

SYNTAX AND COMMANDS TO ADD, RETRIEVE AND DELETE DATA FROM HDFS

Step-1

Adding Files and Directories to HDFS

Before you can run Hadoop programs on data stored in HDFS, you will need to put the data into HDFS first. Let's create a directory and put a file in it. HDFS has a default working directory of /user/$USER, where $USER is your login user name. This directory is not automatically created for you, though, so let's create it with the mkdir command. For the purpose of illustration, we use chuck. You should substitute your user name in the example commands.

hadoop fs -mkdir /user/chuck

hadoop fs -put example.txt

hadoop fs -put example.txt /user/chuck

Step-2

Retrieving Files from HDFS

The Hadoop command get copies files from HDFS back to the local file system. To retrieveexample.txt, we can run the following command:

hadoop fs -cat example.txt

Step-3

Deleting Files from HDFS

hadoop fs -rm example.txt

- Command for creating a directory in hdfs is **"hdfs dfs –mkdir /lendicse"**.
- Adding directory is done through the command **"hdfs dfs –put lendi_english /"**.

Step-4

Copying Data from NFS to HDFS

Copying from directory command is **"hdfs dfs –copy From Local**

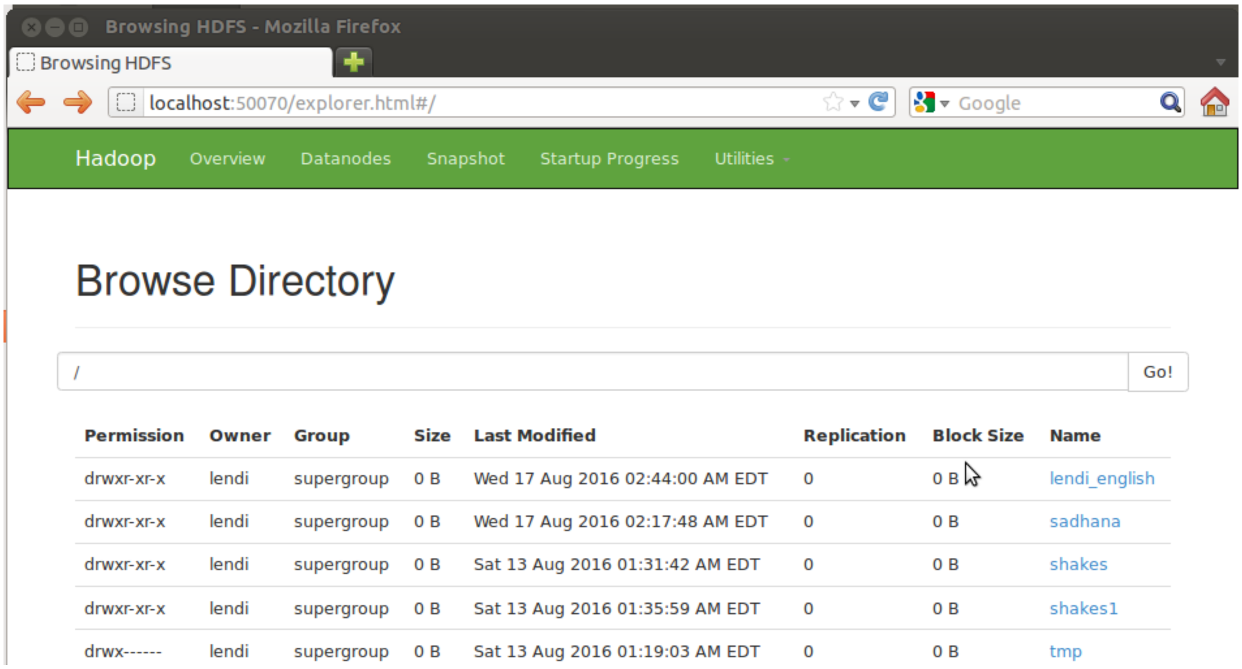
/home/lendi/Desktop/shakes/glossary /lendicse/"

- View the file by using the command **"hdfs dfs –cat /lendi_english/glossary"**
- Command for listing of items in Hadoop is **"hdfs dfs –ls hdfs://localhost:9000/"**.
- Command for Deleting files is **"hdfs dfs –rm r /kartheek"**.

SAMPLE INPUT:

Input as any data format of type structured, Unstructured or Semi Structured

EXPECTED OUTPUT:



RESULT:

Thus the various file management tasks in Hadoop were implemented.